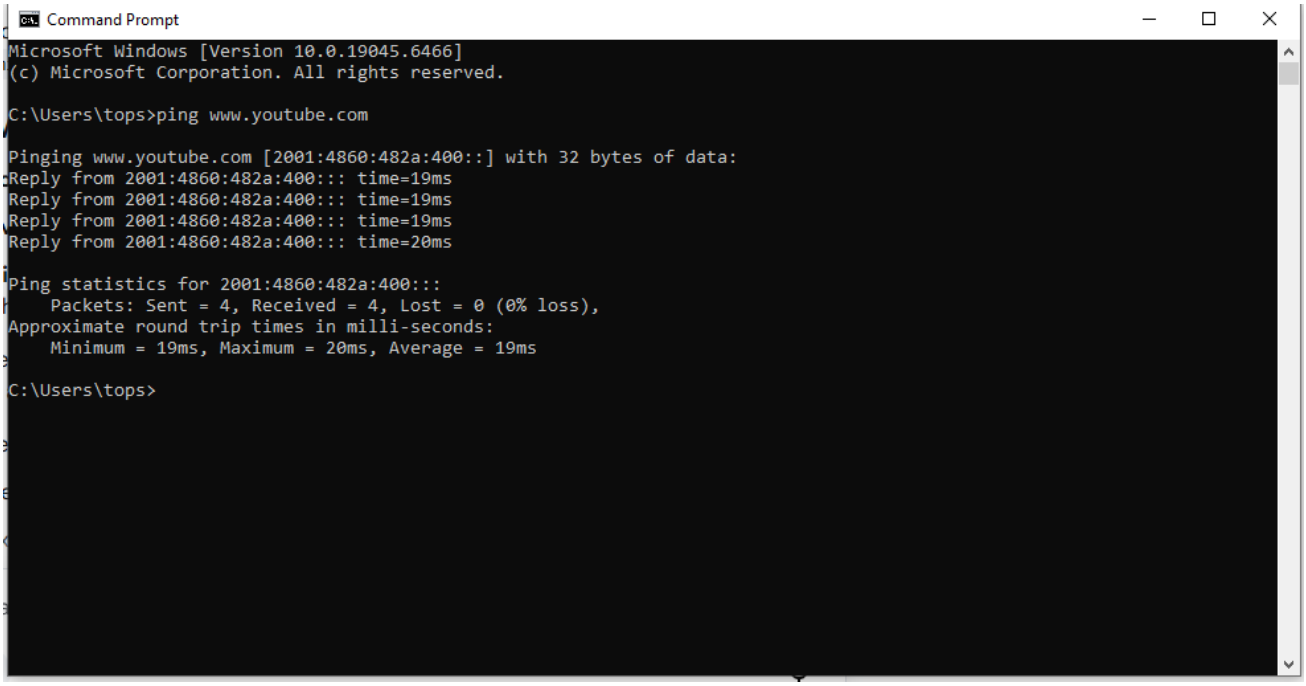
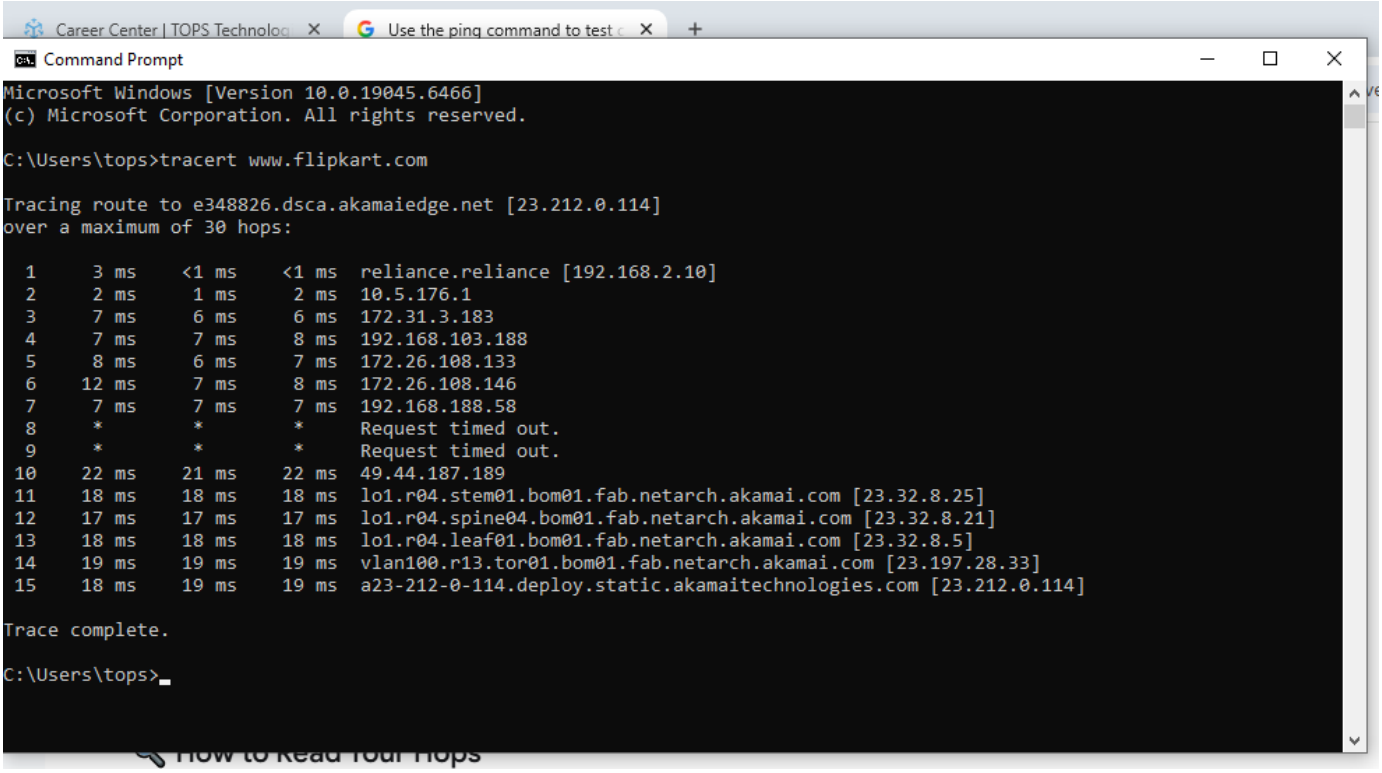
1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.



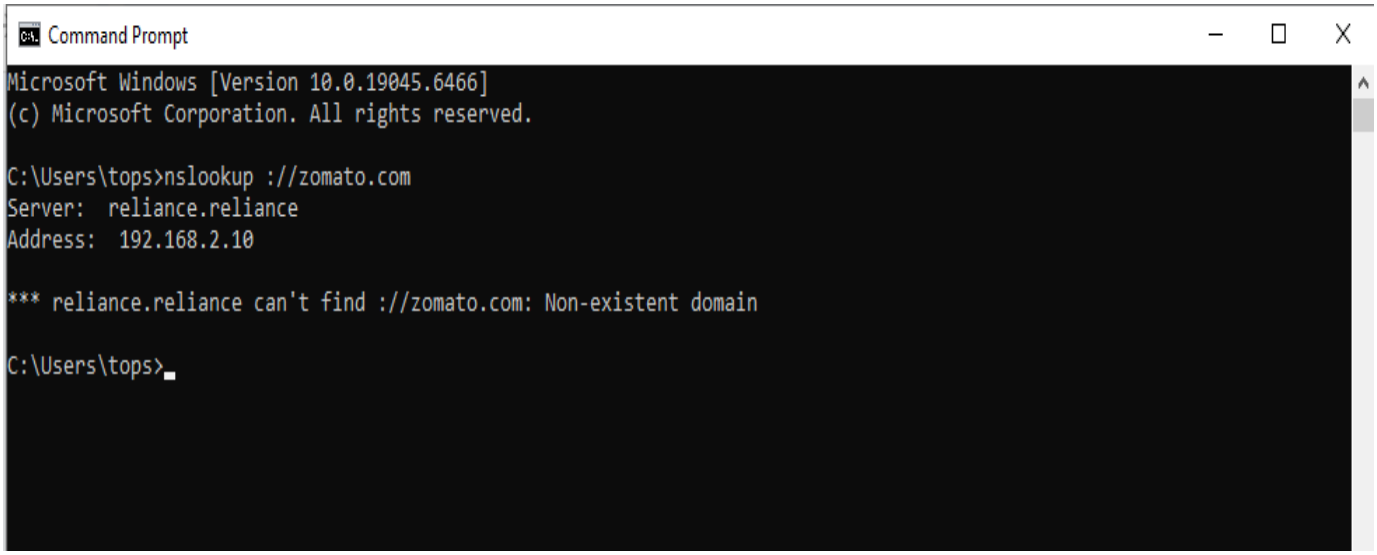
The average response time is 19ms that can be considered excellent connection and highly responsive.

2. Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Hint: Each hop represents a network device your data passes through to reach the destination.



3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.



4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Step 1: Connect Both Devices

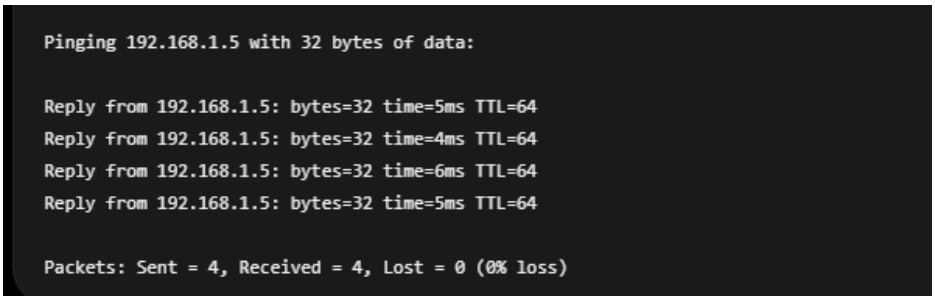
- Connect your laptop and phone to the same Wi-Fi network.
- Find your phone's IP address in **Settings → Wi-Fi → Network Details**.
- Example IP address: 192.168.1.5

Step 2: Open Command Prompt

On Windows, press Win + R, type cmd, and press Enter.

Step 3: Run the Ping Command

ping 192.168.1.5



Result: The device is online and responding to ping requests.

Step 4: Test When the Device Is Offline

Turn off Wi-Fi on your phone and run the same command again.

ping 192.168.1.5

```
Pinging 192.168.1.5 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

5. Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.

Network Troubleshooting Report

1. Ping: Used to check whether a website or device is reachable when it is not loading.
2. **Example:** ping 192.168.1.5 — In the earlier sample, an offline device showed 100% packet loss.
3. **Traceroute:** Used to identify where delays occur along the network path during a slow connection.
4. **Example:** tracert google.com — High response times at a hop may indicate network delays.
5. **NSLookup:** Used to diagnose DNS problems when a website's domain name cannot be resolved.
6. **Example: nslookup google.com —** Displays the IP address associated with the domain if DNS resolution succeeds.
7. **Conclusion:** Ping checks connectivity, Traceroute identifies network path delays, and NSLookup checks DNS resolution.